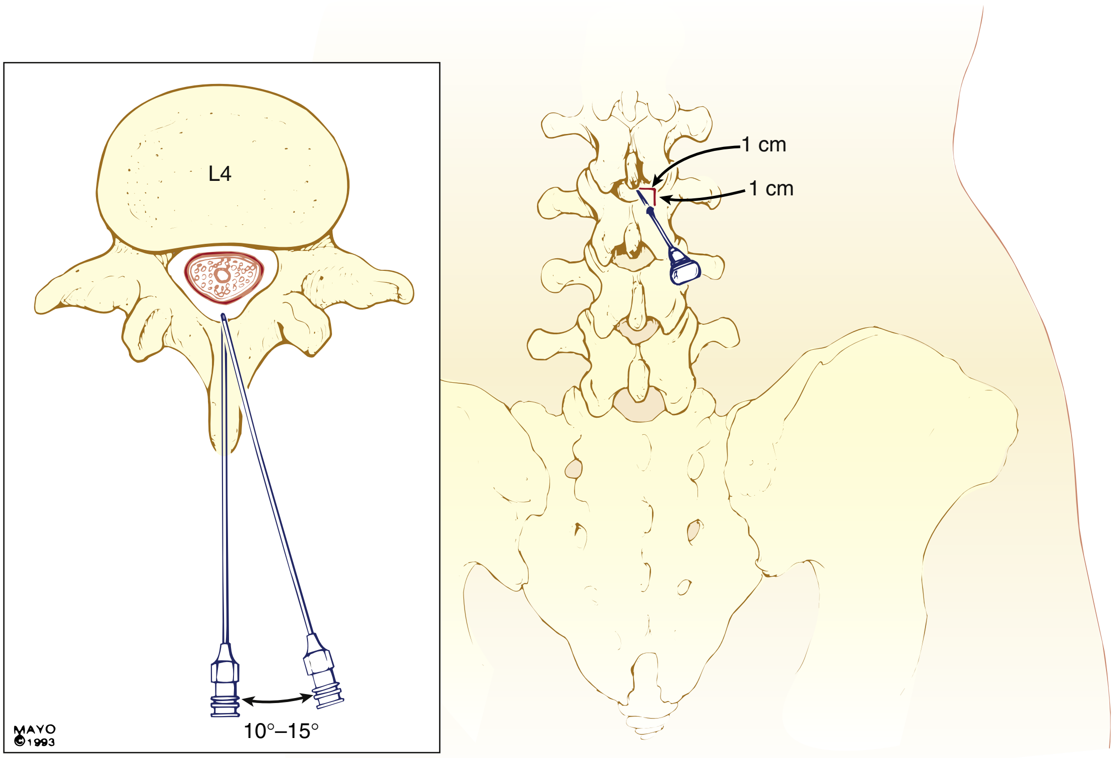

Fig. 17.11 Vertebral anatomy of the midline and paramedian approaches to central neuraxis blocks. The midline approach highlighted in the inset requires anatomic projection in only two planes: sagittal and horizontal. The paramedian approach shown in the inset and in the posterior view requires an additional oblique plane to be considered, although the technique may be easier in patients who are unable to cooperate in minimizing their lumbar lordosis. The paramedian needle is inserted 1 cm lateral and 1 cm caudad to the caudad edge of the more superior vertebral spinous process. The paramedian needle is inserted approximately 15 degrees off the sagittal plane, as shown in the inset. (Courtesy of the Mayo Foundation, Rochester, Minn.)

Monitoring of the Block

Once the spinal anesthetic has been administered, the onset, extent, and quality of the sensory and motor blocks must be assessed while heart rate and arterial blood pressure should be monitored for any resultant sympathetic blockade. Cold sensation and pinprick representing C fibers and A-delta fibers respectively are used most often to assess sensory block. Loss of sensation to cold usually occurs first, verified using an ethyl chloride spray, ice, or alcohol, followed by the loss of sensation to pinprick, verified using a needle that does not pierce the skin.[6] Finally, loss of sensation to touch occurs. The modified Bromage scale (Box 17.1) is most commonly used to measure motor block, although this represents only lumbosacral motor fibers.[53] Ensuring that the level of block using cold or pinprick is two to three segments above the expected level of surgical stimulus, and the presence of a motor block is commonly considered adequate.

Box 17.1 Modified Bromage Scale

0 No motor block
1 Inability to raise extended leg; able to move knees and feet
2 Inability to raise extended leg and move knee; able to move feet
3 Complete block of motor limb

From Brull R, Macfarlane AJR, Chan VWS. Spinal, epidural, and caudal anesthesia. In Miller RD, Cohen NH, Eriksson LI, et al, eds. *Miller's Anesthesia.* 8th ed. Philadelphia: Saunders Elsevier; 2015, Box 56-1.

EPIDURAL ANESTHESIA

Factors Affecting Epidural Block Height

Drug Factors

The volume and total mass of injectate are the most important drug-related factors. As a general principle, 1 to 2 mL of solution should be injected per segment to be

Table 17.4 Factors Affecting Epidural Local Anesthetic Distribution and Block Height

Factors	More Important	Less Important	Not Important
Drug factors	Volume Dose	Concentration	Additives
Patient factors	Elderly age Pregnancy	Weight Height Pressure in adjacent body cavities	
Procedure factors	Level of injection	Patient position	Speed of injection Needle orifice direction

Visser WA, Lee RA, Gielen MJM. Factors affecting the distribution of neural blockade by local anesthetics in epidural anesthesia and a comparison of lumbar versus thoracic epidural anesthesia. *Anesth Analg*. 2008;107(2):708-721.

blocked. Bicarbonate, epinephrine, and opioids influence onset, quality, and duration of analgesia and anesthesia, and these do not affect spread (Table 17.4).

Patient Factors

Age can influence epidural block height. Up to 40% less volume may be required in thoracic epidurals in the elderly, possibly because of decreased leakage of local anesthetic through intervertebral foramina, decreased compliance of the epidural space, or an increased sensitivity of the nerves (also see Chapter 35). Only the extremes of patient height influence local anesthetic spread in the epidural space. Weight is not well correlated with block height. Less local anesthetic is required to produce the same epidural spread of anesthesia in pregnant patients, in part because of engorgement of epidural veins secondary to increased abdominal pressure (also see Chapter 33). Continuous positive airway pressure increases the height of a thoracic epidural block.

Procedure Factors

The level of injection is the most important procedural-related factor that affects epidural block height. In the upper cervical region, spread of injectate is mostly caudal, in the midthoracic region spread is equally cephalad and caudal, and in the low thoracic region spread is primarily cephalad.[54] After a lumbar epidural, spread is more cephalad than caudal. Some studies suggest that the total number of segments blocked is less in the lumbar region compared with thoracic levels for a given volume of injectate. Patient position affects lumbar epidural injections, with preferential spread and faster onset to the dependent side in the lateral decubitus position. The sitting and supine positions do not affect epidural block height. However, the head-down tilt position does increase spread in obstetric patients. Needle bevel direction and speed of injection do not appear to influence the spread of a bolus injection.

Pharmacology

Local anesthetics for epidural use may be classified into short-, intermediate-, and long-acting drugs. A single bolus dose of local anesthetic can provide surgical anesthesia ranging from 45 minutes up to 4 hours depending on the type administered and the use of any additives (Table 17.5). Most commonly, however, an epidural catheter is left in situ so that anesthesia or analgesia can be extended indefinitely.

Short-Acting and Intermediate-Acting Local Anesthetics

Procaine is not commonly used because the resultant block can be unreliable and of poor quality.

Chloroprocaine is available in 2% and 3% preservative-free solutions. Prior to preservative-free preparations, large volumes of chloroprocaine had been associated with deep, burning lumbar back pain.[55] This was thought to be secondary to the ethylenediaminetetraacetic acid that chelated calcium and caused a localized hypocalcemia.

Articaine is not widely used for epidural anesthesia and has not been studied extensively.

Lidocaine is available in 1% and 2% solutions. Unlike spinal anesthesia, TNS is not commonly associated with epidural lidocaine.

Prilocaine is available in 2% and 3% solutions. The 2% solution produces a sensory block with minimal motor block. In large doses, prilocaine is associated with methemoglobinemia.

Mepivacaine is available as 1%, 1.5%, and 2% preservative-free solutions. The 2% preparation has an onset time similar to that for lidocaine of approximately 15 minutes, but a slightly longer duration (up to 200 minutes with epinephrine).

Long-Acting Local Anesthetics

Tetracaine is not widely used because of unreliable block height and, in larger doses, systemic toxicity.

Bupivacaine is available in 0.25%, 0.5%, and 0.75% preservative-free solutions. More dilute concentrations such

Table 17.5 Comparative Onset Times and Analgesic Durations of Local Anesthetics Administered Epidurally in 20- to 30-mL Volumes

			Duration (min)	
Drug	**Conc. (%)**	**Onset (min)**	**Plain**	**1 : 200,000 Epinephrine**
2-Chloro-procaine	3	10-15	45-60	60-90
Lidocaine	2	15	80-120	120-180
Mepivacaine	2	15	90-140	140-200
Bupivacaine	0.5-0.75	20	165-225	180-240
Etidocaine	1	15	120-200	150-225
Ropivacaine	0.75-1.0	15-20	140-180	150-200
Levobupivacaine	0.5-0.75	15-20	150-225	150-240

Data from Cousins MJ, Bromage PR. Epidural neural blockade. In Cousins MJ, Bridenbaugh PO, eds. *Neural Blockade in Clinical Anesthesia and Management of Pain*. Philadelphia: JB Lippincott; 1988:255; Brown DL. Spinal, epidural, and caudal anesthesia. In Miller RD, Cohen NH, Eriksson LI, et al, eds. *Miller's Anesthesia*. 7th ed. Philadelphia: Saunders Elsevier; 2010:1611-1638.

as 0.125% to 0.25% can be used for analgesia. However, disadvantages include cardiac and central nervous system toxicity and potential motor block from larger doses.

Levobupivacaine administered epidurally has the same clinical characteristics as bupivacaine and is less cardiotoxic. Liposomal bupivacaine is not licensed for epidural use.

Ropivacaine is available in 0.2%, 0.5%, 0.75%, and 1.0% preservative-free preparations. It is associated with a superior safety profile compared with bupivacaine, with a higher seizure threshold and less cardiotoxicity.

Epidural Additives

Vasoconstrictors

Epinephrine reduces vascular absorption of local anesthetics in the epidural space. The effect is greatest with lidocaine,[56] mepivacaine, and chloroprocaine (up to 50% prolongation); less with bupivacaine and levobupivacaine; and limited with ropivacaine, which already has intrinsic vasoconstrictive properties (see Table 17.5). Epinephrine itself may also have some analgesic benefits owing to absorption into CSF, and dorsal horn α_2-receptor activation. Phenylephrine is used less widely and is less effective than epinephrine.

Opioids

Opioids synergistically enhance the analgesic effects of epidural local anesthetics, without prolonging motor block. A combination of local anesthetic and opioid reduces the dose-related adverse effects of each drug independently. Opioid-related side effects are dose-dependent and there appears to be a therapeutic ceiling effect above which only side effects increase. Opioids may also be used alone. Epidural opioids cross the dura and arachnoid membrane to reach the CSF and spinal cord dorsal horn. Lipophilic opioids, such as fentanyl and sufentanil, partition into epidural fat and therefore are found in lower concentrations in CSF than hydrophilic opioids, such as morphine and hydromorphone. Fentanyl and sufentanil are also readily absorbed into the systemic circulation, which may be the principal analgesic mechanism.

Epidural morphine can be administered as a bolus (duration of up to 24 hours) or continuously. The optimal bolus analgesic dose that minimizes side effects is 2.5 to 3.75 mg.[57] Hydromorphone is more hydrophilic than fentanyl but more lipophilic than morphine, and has a duration of 18 hours. Epidural fentanyl and sufentanil have a faster onset but a shorter duration (only 2 to 3 hours). Diamorphine is available in the United Kingdom. Depodur is an extended-release liposomal formulation of morphine used as a single-shot lumbar epidural dose, potentially avoiding issues of continuous local anesthetic infusions and indwelling catheters.

α_2-Agonists

Epidural clonidine can prolong sensory block to a longer extent than motor block and reduces both epidural local anesthetic and opioid requirements. Clonidine may also reduce immune stress and cytokine response, but side effects include hypotension, bradycardia, dry mouth, and sedation. Dexmedetomidine can reduce intraoperative anesthetic requirements, improve postoperative analgesia, and prolong both sensory and motor block.

Other Drugs

Ketamine, neostigmine, midazolam, tramadol, dexamethasone, and droperidol have all been studied but are not commonly used.

Table 17.6 Suggested Epidural Insertion Sites for Common Surgical Procedures

Nature of Surgery	Suggested Level of Insertion	Remarks
Hip surgery Lower extremity Obstetric analgesia	Lumbar L2-L5	
Colectomy, anterior resection Upper abdominal surgery	Lower thoracic T6-T8	Spread more cranial than caudal
Thoracic	T2-T6	Midpoint of surgical incision

Modified from Visser WA, Lee RA, Gielen MJM. Factors affecting the distribution of neural blockade by local anesthetics in epidural anesthesia and a comparison of lumbar versus thoracic epidural anesthesia. *Anesth Analg*. 2008;107(2):708-721.

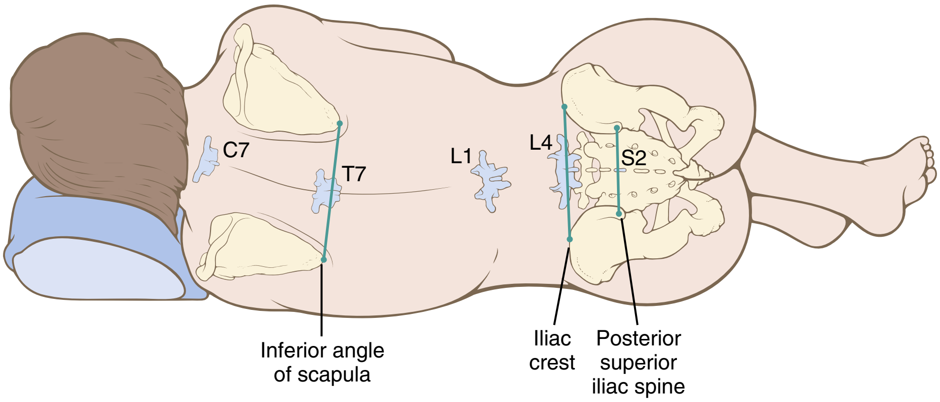

Fig. 17.12 Surface landmarks are a guide to the vertebral level. (From Brown DL, ed. *Atlas of Regional Anesthesia*. Philadelphia: WB Saunders; 1992.)

Carbonation and Bicarbonate

Both carbonation of the solution and adding bicarbonate increase the solution pH, and therefore the nonionized free-base proportion of local anesthetic. Although this may theoretically increase the speed of onset and quality of the block by producing more rapid intraneural diffusion and more rapid penetration of connective tissue surrounding the nerve trunk, data suggest that there are no clinical advantages for carbonated solutions.[58,59]

Technique

Preparation

Patient preparation as previously described for spinal anesthesia must equally be applied to epidural anesthesia, namely consent, monitoring, resuscitation equipment, and intravenous access. Sterility is arguably even more important than spinal anesthesia because a catheter is often left in situ. The nature and the duration of surgery must be understood so that the epidural may be inserted at the appropriate level (Table 17.6) and the appropriate drugs chosen.[8] The risks and benefits will vary depending on the severity of patient comorbid conditions. Tuohy needles are most commonly used (see Fig. 17.10). They are usually 16 to 18 G and have a shaft marked in 1-cm intervals with a 15- to 30-degree curved, blunt Huber needle tip designed to both reduce the risk of accidental dural puncture and guide the catheter cephalad. The catheter is made of a flexible, calibrated, radiopaque plastic with either a single end hole or multiple side orifices near the tip. Multiple orifices can improve analgesia but may increase epidural vein cannulation in parturients.

Position

The sitting and lateral decubitus positions necessary for epidural puncture are the same as those for spinal anesthesia, and success rates are comparable. As with spinal anesthesia, epidurals are ideally performed with the patient awake.[21]

Projection and Puncture

Important surface landmarks include the intercristal line (corresponding to the L4-L5 interspace), the inferior angle of the scapula (corresponding to the T7 vertebral body), the root of the scapular spine (T3), and the vertebra prominens (C7) (Fig. 17.12). Ultrasonography may be useful to identify the correct thoracic space.

A variety of different needle approaches exist: midline, paramedian, modified paramedian (Taylor approach), and caudal. A midline approach, in which the angle of approach is only slightly cephalad, is commonly chosen for lumbar and low thoracic approaches. In the midthoracic region, the approach should be more cephalad because of the significant downward angulation of the spinous processes (see Fig. 17.5). The needle should be advanced in a controlled fashion with the stylet in place through the supraspinous ligament and into the interspinous ligament, at which point the stylet can be removed and the syringe attached. This method may increase the chance of a false loss-of-resistance, possibly because of defects in the interspinous ligament.

Air or saline (or a combination) is commonly used to detect a loss-of-resistance when identifying the epidural space (Fig. 17.13). Each involves intermittent (for air) or constant (for saline) gentle pressure applied to the bulb of the syringe with the dominant thumb while the needle is advanced with the nondominant hand. Usually the ligamentum flavum is identified as a tougher structure with increased resistance, and when the epidural space is subsequently entered, the pressure applied to the syringe plunger allows the solution to flow without resistance into the epidural space. Air is likely less reliable in identifying the epidural space, results in a possible chance of incomplete block, and may cause both pneumocephalus (which can result in headaches) and even venous air embolism in rare cases. Nevertheless, adverse outcomes in obstetric patients do not vary when air versus saline was studied.[60] Fluid inserted through the epidural needle before catheter insertion can also reduce the risk of epidural vein cannulation by the catheter.[61] However, saline may make it more difficult to detect an accidental dural puncture.

With the hanging-drop technique, a drop of solution such as saline is placed within the hub of the needle after the needle is placed in the ligamentum flavum. When the needle tip reaches the epidural space, the solution is "sucked in" as a result of subatmospheric pressure inside the epidural space.

When a lumbar midline approach is used, the depth from skin to the ligamentum flavum in most (80%) patients is between 3.5 and 6 cm. Ultrasonography can predict this depth before needle insertion. When the epidural space is identified, the depth should be noted, the syringe removed, and a catheter gently threaded to leave 4 to 6 cm in the space. Less than 4 cm in length in the epidural space may increase the risk of catheter dislodgement and inadequate analgesia. Threading more catheter increases the likelihood of catheter malposition or complications.[62] The Tsui test can be used to confirm catheter tip location using a special electrically conducting catheter.[63] This stimulates the spinal nerve roots with a low electrical current resulting in twitches of the corresponding muscles.

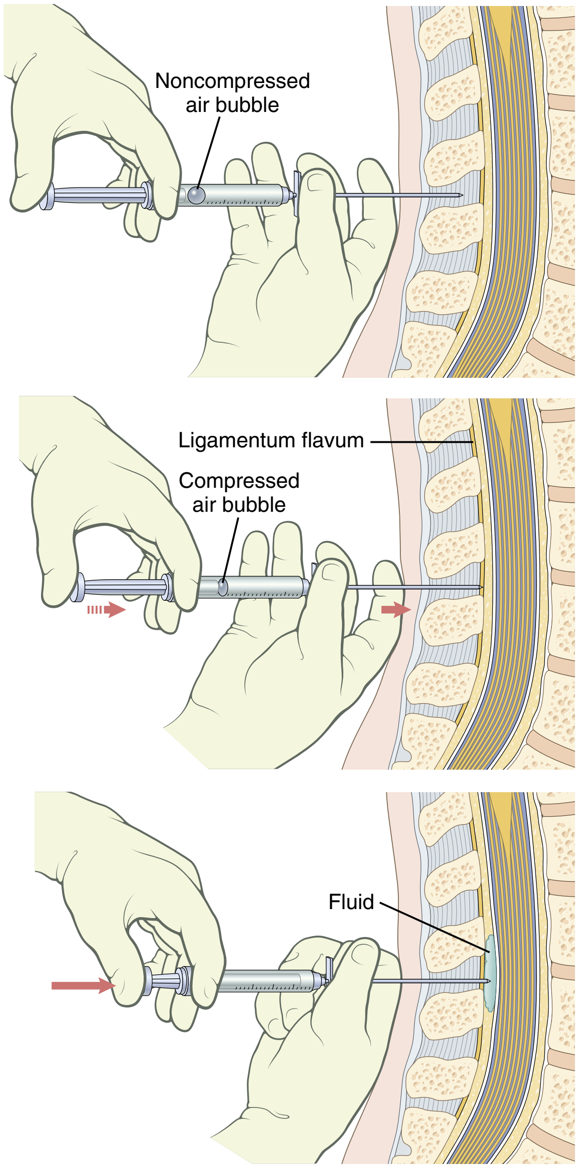

Fig. 17.13 Loss-of-resistance technique. The needle is inserted into the ligamentum flavum, and a syringe containing saline and an air bubble is attached to the hub. After compression of the air bubble is obtained by applying pressure on the syringe plunger, the needle is carefully advanced until its entry into the epidural space is confirmed by the characteristic loss of resistance to syringe plunger pressure, and the fluid enters the space easily. (From Afton-Bird G. Atlas of regional anesthesia. In Miller RD, ed. *Miller's Anesthesia*. Philadelphia: Elsevier; 2005.)

Paramedian Approach

The paramedian approach is particularly useful in the mid- to high-thoracic region, where the angulation of the spine is steeper and the spaces narrower. The needle should be inserted 1 to 2 cm lateral to the inferior tip of the spinous process corresponding to the vertebra above the desired

III

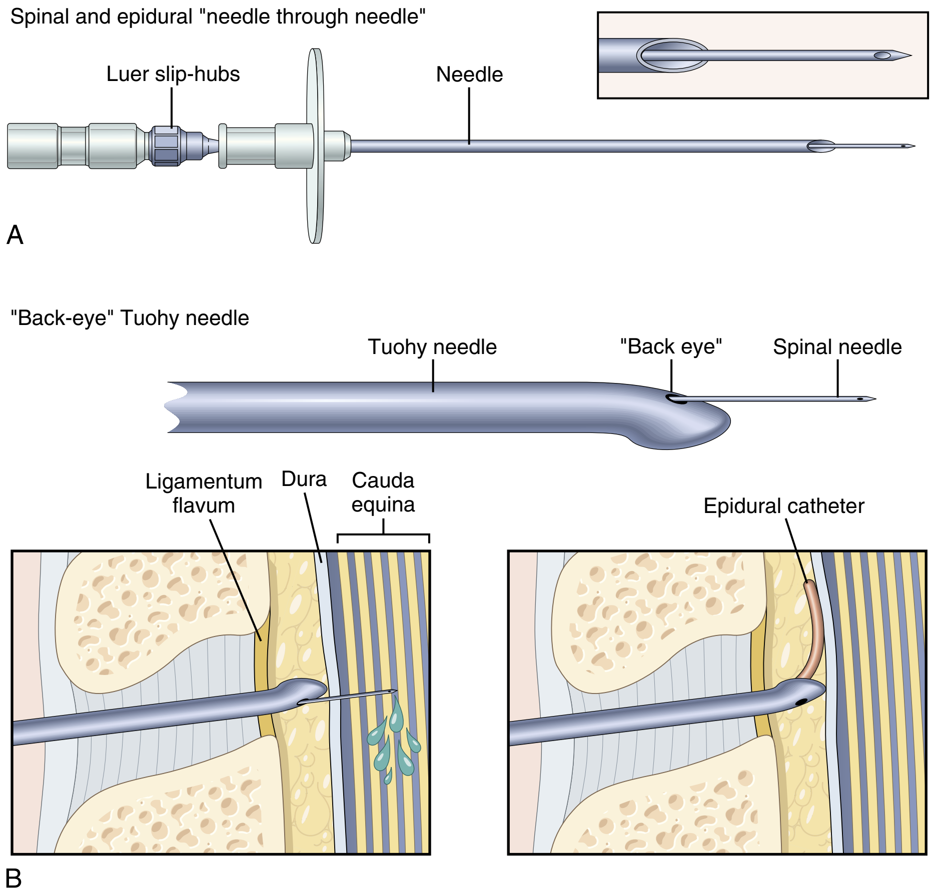

Fig. 17.14 (A) A spinal needle and epidural needle are used for the combined spinal-epidural technique. (B) Tuohy needle with a "back eye" that permits placement of the spinal needle directly into the subarachnoid space *(left panel)* and subsequent threading of the epidural catheter into the epidural space after removal of the spinal needle. (Modified from Veering BT, Cousins MJ. Epidural neural blockade. In Cousins MJ, Bridenbaugh PO, Carr DB, Horlocker TT, eds. *Neural Blockade in Clinical Anesthesia and Management of Pain.* Philadelphia: Lippincott-Raven; 2009:241-295.)

interspace. The needle is then advanced horizontally until the lamina is reached and then redirected medially and cephalad to enter the epidural space. The Taylor approach is a modified paramedian approach via the L5-S1 interspace, which may be useful in trauma patients who cannot tolerate or are not able to maintain a sitting position. The needle is inserted 1 cm medial and 1 cm inferior to the posterior superior iliac spine and is angled medially and cephalad at a 45- to 55-degree angle. Before initiating an epidural local anesthetic infusion, a test dose may be administered. The purpose of this is to exclude intrathecal or intravascular catheter placement.

COMBINED SPINAL-EPIDURAL ANESTHESIA

CSE anesthesia allows the flexibility of a rapid-onset spinal block while the epidural catheter allows anesthesia or analgesia to be extended as the spinal resolves. This is particularly useful in obstetrics (also see Chapter 33). Another advantage is the ability to administer a low dose of intrathecal local anesthetic, and, if necessary, use the epidural catheter to extend the block. Adding either local anesthetic or saline alone to the epidural space via the catheter compresses the dural sac and increases the block height. This latter technique of epidural volume extension (EVE) allows smaller doses of intrathecal local anesthetic to be used while significantly hastening motor recovery.[64] This sequential technique also provides greater hemodynamic stability for high-risk patients.

Technique

Most commonly the epidural needle is placed first, followed by either a "needle through needle" technique using specially available kits (Fig. 17.14) or an altogether separate spinal needle insertion at either the same or

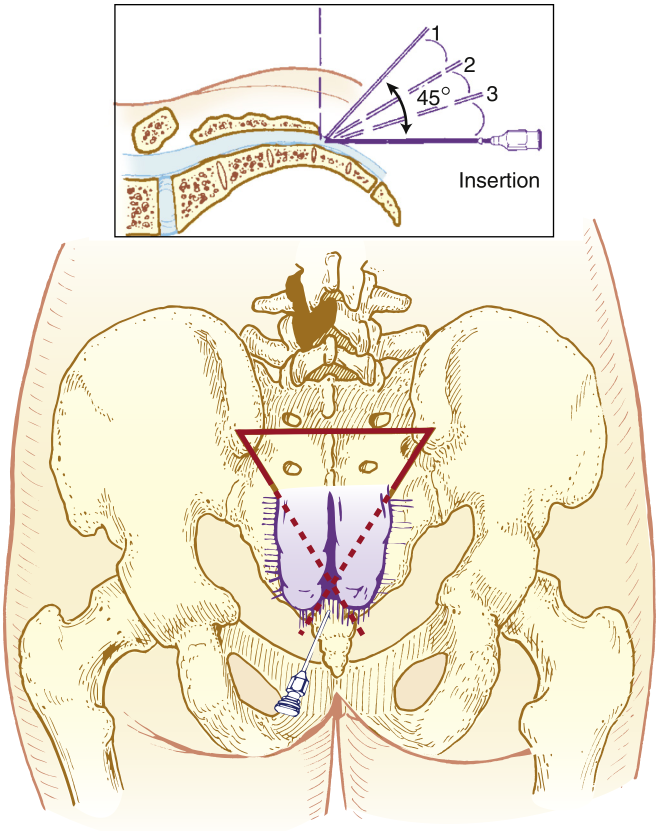

Fig. 17.15 Caudal technique. Palpating fingers locate the sacral cornua by using the equilateral triangle. Needle insertion is completed by insertion and withdrawal in a stepwise fashion (inset, so-called "1-2-3 insertion") until the needle can be advanced into the caudal canal and the solution can be injected easily (without creation of a subcutaneous "lump" of fluid). (From Brull R Macfarlane AJR, Chan VWS. Spinal, epidural, and caudal anesthesia. In Miller RD, Cohen NH, Eriksson LI, et al, eds. *Miller's Anesthesia.* 8th ed. Philadelphia: Saunders Elsevier; 2015:Fig. 56-10.)

different interspace. The separate needle insertion technique[65] has the advantage of being able to confirm that the epidural catheter is functional before spinal anesthesia is administered but does theoretically risk shearing the in situ epidural catheter.

CAUDAL ANESTHESIA

Caudal anesthesia is popular in pediatric anesthesia (also see Chapter 34). In adults, caudal anesthesia is unpredictable when upper abdominal or thoracic spread is required. Therefore its indications in adults are the same as those for lumbar epidural anesthesia. It is most useful when sacral anesthetic spread is desired (e.g., perineal, anal, rectal procedures), where a spinal surgery scar may prevent a lumbar anesthetic technique, or more commonly, in chronic pain and cancer pain management (also see Chapter 44). Both fluoroscopy and ultrasonography can help guide correct needle placement.

Pharmacology

The local anesthetics used are similar to those described for epidural anesthesia and analgesia. In adults, approximately twice the lumbar epidural dose is required to achieve a similar block with the caudal approach.

Technique

Patient preparation as described before applies equally to caudal anesthesia. The prone position, lateral decubitus position, and the knee-chest position may be used. Caudal anesthesia requires identification of the sacral hiatus. The sacrococcygeal ligament (i.e., extension of ligamentum flavum) overlies the sacral hiatus, between the two sacral cornua. The posterior superior iliac spines should be located, and by using the line between them as one side of an equilateral triangle, the location of the sacral hiatus can be approximated (Fig. 17.15).

After the sacral hiatus is identified, local anesthetic is infiltrated and the caudal needle (or Tuohy needle if a catheter is to be placed) is inserted at an angle of approximately 45 degrees to the sacrum. A decrease in resistance to needle insertion should be appreciated as the needle enters the caudal canal. The needle is advanced until bone (i.e., the dorsal aspect of the ventral plate of the sacrum) is contacted and then slightly withdrawn, and then redirected so that the angle of insertion relative to the skin surface is decreased. In male patients, this angle is almost parallel to the coronal plane; in female patients, a slightly steeper angle (15 degrees) is necessary. During redirection of the needle, loss-of-resistance is sought to confirm entry into the epidural space, and the needle advanced no more than approximately 1 to 2 cm into the caudal canal. In adults, the tip should never be advanced beyond the S2 level (approximately 1 cm inferior to the posterior superior iliac spine), which is the level to which the dural sac extends. Additional advancement increases the risk of dural puncture and unintentional intravascular cannulation. Before injection of any drugs, aspiration to see if CSF can be withdrawn should be performed and a test dose administered to exclude intravascular or intrathecal placement.

COMPLICATIONS

Clear distinction should be made between the physiologic effects of the neuraxial technique and complications, which imply some harm to the patient.[66] The material risks associated with neuraxial anesthesia must be intimately understood and respected because catastrophic injury is not unknown.

Neurologic

Serious neurologic complications associated with neuraxial anesthesia are rare.

Paraplegia

The frequency is reported to be approximately 0.1 per 10,000.[67] The mechanism of such a severe injury is likely multifactorial. Direct needle trauma to the spinal cord is possible, but the intrathecal injectate can be neurotoxic. In the early 1980s, several patients developed adhesive arachnoiditis, cauda equina syndrome, or permanent paresis possibly related to a combination of low pH and the antioxidant sodium bisulfite preservative used in early (and discontinued) preparations of the short-acting ester local anesthetic chloroprocaine.[68]

Profound hypotension or ischemia of the spinal cord can also be an important contributing factor. Anterior spinal artery syndrome, characterized by potentially irreversible, painless loss of motor and sensory function with sparing of proprioception was described earlier.

Cauda Equina Syndrome

The rate of cauda equina syndrome is approximately 0.1 per 10,000 and invariably results in permanent neurologic deficit.[69] The lumbosacral roots of the spinal cord may be particularly vulnerable to direct exposure of large doses of local anesthetic, whether it is administered as a single injection of relatively highly concentrated local anesthetic (e.g., 5% lidocaine) or prolonged exposure to a local anesthetic through a (small gauge in particular as described earlier) continuous catheter.

Epidural Hematoma

Bleeding within the vertebral canal can cause ischemic compression of the spinal cord and lead to permanent neurologic deficit if not recognized and evacuated expeditiously. Many risk factors have been associated with the development of an epidural hematoma, including difficult or traumatic needle or catheter insertion, coagulopathy, elderly age, and female gender.[70] Radicular back pain, prolonged blockade longer than the expected duration of the neuraxial technique, and bladder or bowel dysfunction are features commonly associated with a space-occupying lesion within the vertebral canal and should prompt magnetic resonance imaging on an urgent basis. The rate is about 0.07 per 10,000.[67]

Nerve Injury

The current conclusion is that epidural (including CSE) anesthesia is associated with a more frequent rate of radiculopathy or peripheral neuropathy compared with spinal anesthesia,[69] and that neuraxial anesthesia performed in adults for the purposes of perioperative anesthesia or analgesia is associated with an increased likelihood of neurologic complications compared with that performed in the obstetric, pediatric, and chronic pain settings.[67] The rate of permanent nerve injury is around 0.1 per 10,000.[67] Radicular pain or paresthesia occurring during the procedure is a risk factor.

Post–Dural Puncture Headache

This headache is relatively common and results from unintentional or intentional puncture of the dura membrane. The incidence is approximately 1% in spinal anesthesia and is minimized by using smaller gauge, noncutting tip spinal needles and orientating the needle bevel parallel with the axis of the spine. In obstetrics, accidental dural puncture may occur in around 1.5% of patients with between 52% and 80% of these patients subsequently developing a post–dural puncture headache.[71] Additional risk factors for post–spinal puncture headache are listed in Box 17.2.

The loss of CSF through the dura may cause traction on pain-sensitive intracranial structures as the brain loses support and sags (Fig. 17.16). Perhaps the loss of CSF initiates compensatory yet painful intracerebral vasodilation to offset the decreased intracranial pressure.[72] The characteristic feature of a post–dural puncture headache is a frontal or occipital headache that worsens with the upright or seated posture and is relieved by lying supine. Associated symptoms can include nausea, vomiting, neck pain, dizziness, tinnitus, diplopia, hearing loss, cortical blindness, cranial nerve palsies, and even seizures. Symptoms usually begin within 3 days of the procedure, and 66% start within the first 48 hours. Spontaneous resolution usually occurs within 7 days in the majority (72%) of cases, whereas 87% of cases resolve by 6 months.

Box 17.2 Relationships Among Variables and Post Dural Puncture Headache

Factors That Can Increase the Incidence of Headache After Spinal Puncture

- Age: Younger, more frequent
- Sex: Females > males
- Needle size: larger > smaller
- Needle bevel: less when the needle bevel is placed in the long axis of the neuraxis
- Pregnancy: more when pregnant
- Dural punctures: more with multiple punctures

Factors That Do Not Increase the Incidence of Headache After Spinal Puncture

- Continuous spinal infusion
- Timing of ambulation

From Brull R, Macfarlane AJR, Chan VWS. Spinal, epidural, and caudal anesthesia. In Miller RD, Cohen NH, Eriksson LI, et al, eds. *Miller's Anesthesia.* 8th ed. Philadelphia: Saunders Elsevier; 2015:Box 56-2.

Conservative management for post–dural puncture headache includes supine positioning, hydration, caffeine, and oral analgesics. The use of sumatriptan has varying effects. Epidural blood patch is the definitive therapy for post–dural puncture headache[73] with a single patch resulting in a 90% initial improvement rate and persistent resolution of symptoms in 61% to 75% of cases. A blood patch is ideally performed 24 hours after dural puncture and after the development of classic post–dural puncture headache symptoms. Prophylactic epidural blood patching is not efficacious. It is recommended inserting the blood patch needle at or caudad to the level of the dural puncture, with 20 mL of blood being a reasonable starting target volume.[74] A second epidural blood patch may be performed 24 to 48 hours if the first is ineffective.

Transient Neurologic Symptoms

TNS are characterized by bilateral or unilateral pain in the buttocks radiating to the legs or, less commonly, isolated buttock or leg pain. Symptoms occur within 24 hours of the resolution of an otherwise uneventful spinal anesthetic and are not associated with any neurologic deficits or laboratory abnormalities. Pain can be mild or severe but typically resolves spontaneously in less than 1 week. TNS are more likely after intrathecal lidocaine and mepivacaine and are far less frequent with bupivacaine.[75] The phenomenon is related to the concentration of lidocaine, the addition of dextrose or epinephrine, and solution osmolarity. TNS are less commonly associated with

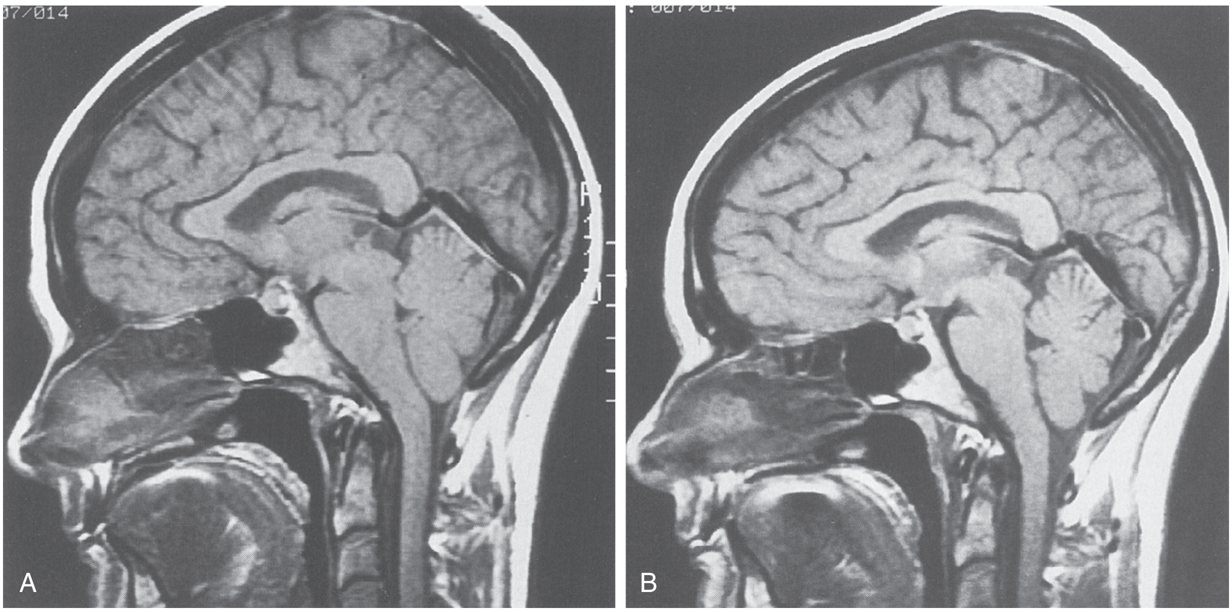

Fig. 17.16 Anatomy of a "low-pressure" headache. (A) A T1-weighted sagittal magnetic resonance image demonstrates a "ptotic brain" manifested as tonsillar herniation below the foramen magnum, forward displacement of the pons, absence of the suprasellar cistern, kinking of the chiasm, and fullness of the pituitary gland. (B) A comparable image of the same patient after an epidural blood patch and resolution of the symptoms demonstrates normal anatomy. (From Drasner K, Swisher JL. In Brown DL, ed. *Regional Anesthesia and Analgesia.* Philadelphia: WB Saunders; 1996.)

epidural procedures. The risk is also more likely in the lithotomy position for surgery. Nonsteroidal antiinflammatory drugs are the first line of treatment, but opioids may be required.

Cardiovascular

Hypotension

Hypotension is more likely with peak block height higher than or equal to T5, at an age of 40 years or older, baseline systolic blood pressure less than 120 mm Hg, combined spinal and general anesthesia, spinal puncture at or above the L2-L3 interspace, and the addition of phenylephrine to the local anesthetic. Hypotension is also independently associated with chronic alcohol consumption, history of hypertension, body mass index (BMI), and the urgency of surgery.[76] Nausea is a common symptom of hypotension in the setting of neuraxial anesthesia; other symptoms include vomiting, dizziness, and dyspnea.

Bradycardia

The mechanism has been described earlier but factors that may increase the likelihood of exaggerated bradycardia include baseline heart rate less than 60 beats/min, age younger than 37 years, male gender, nonemergency status, β-adrenergic blockade, and prolonged duration of surgery.

Cardiac Arrest

This event is rare, and the cause of sudden cardiac arrest after spinal anesthesia is not understood. Hypoxemia and oversedation may be a factor in the severe bradycardia and asystole that can occur suddenly during well-conducted spinal anesthesia. Curiously, these rare events are associated with spinal anesthesia rather than epidural techniques.

Respiratory

The risk of respiratory depression associated with neuraxial opioids is dose-dependent, with a reported frequency that approaches 3% after the administration of 0.8 mg of intrathecal morphine.[77] Respiratory depression may stem from rostral spread of opioids within the CSF to the chemosensitive respiratory centers in the brainstem. With lipophilic anesthetics, respiratory depression is generally an early phenomenon occurring within the first 30 minutes (and has not been reported after 2 hours), whereas with intrathecal morphine late respiratory depression may occur up to 24 hours after injection. Respiratory monitoring for the first 24 hours after the administration of intrathecal morphine is recommended. Patients with sleep apnea can be especially sensitive and considerable caution must be exercised in this group.[78] Older patients also have a more frequent risk of respiratory depression, and therefore, the dose of neuraxial opioids should be reduced (also see Chapter 35). Coadministration of systemic sedatives also increases this risk.

Infection

Bacterial meningitis and epidural abscess are rare but potentially catastrophic complications. Staphylococcal infections arising from organisms on the patient's skin are one of the most common epidural-related infections, whereas oral bacteria such as *S. viridans* are a common cause of infection after spinal anesthesia. The presence of a concomitant systemic infection, diabetes, immunocompromised states, and prolonged maintenance of an epidural (or spinal) catheter are risk factors. The rate of serious neuraxial infection is less than 0.3 per 10,000[79] for spinal anesthesia, whereas infectious complications after epidural techniques may be at least twice as common.[67] Obstetric patients are less likely to develop deep infections related to epidural analgesia. Chlorhexidine in an alcohol base is the most effective antiseptic for the purposes of neuraxial techniques.

Backache

There is no association between epidural analgesia and new-onset back pain up to 6 months postpartum.

Nausea and Vomiting

Nausea and vomiting may be secondary to either direct exposure of the chemoreceptor trigger zone in the brain to emetogenic drugs (e.g., opioids), hypotension, or gastrointestinal hyperperistalsis secondary to unopposed parasympathetic activity. Nausea or vomiting after spinal anesthesia is more likely with the addition of phenylephrine or epinephrine to the local anesthetic, peak block height more than or equal to T5, baseline heart rate more rapid than 60 beats/min, use of procaine, history of motion sickness, and the development of hypotension during spinal anesthesia. Intrathecal morphine has the highest risk of opioid-induced nausea or vomiting, whereas fentanyl and sufentanil carry the lowest risk.[80] Again these side effects are dose-dependent. Using less than 0.1 mg morphine reduces the risk, without compromising the analgesic effect.

Urinary Retention

Urinary retention can occur in as many as one third of patients after neuraxial anesthesia. Local anesthetic blockade of the S2, S3, and S4 nerve roots inhibits urinary function as the detrusor muscle is weakened. Neuraxial opioids can further complicate urinary function by suppressing detrusor contractility and reducing the sensation of urge.[81] Spontaneous return of normal bladder function is expected once the sensory level decreases to less than S2-S3. Along with male gender and age, intrathecal morphine has also been linked to urinary retention after neuraxial anesthesia.[81]

Pruritus

Pruritus can be distressing and is the most common side effect related to the intrathecal administration of opioids with rates between 30% and 100%.[82] It is not dependent on the type or dose of opioid administered, although reducing the dose can reduce the likelihood of pruritus. Naloxone, naltrexone, or the partial agonist nalbuphine can be used for treatment. Ondansetron and propofol are also useful therapies.

Shivering

The rate of shivering is as frequent as 55%[83] and is more related to epidural than spinal anesthesia. One postulated cause is the relatively cold temperature of the epidural injectate, which can affect the thermosensitive basal sinuses. The addition of neuraxial opioids, specifically fentanyl and meperidine, reduces the likelihood of shivering.[83] Prewarming the patient with a forced air warmer and avoiding the administration of cold epidural and intravenous fluids can reduce the incidence of shivering.

Complications Unique to Epidural Anesthesia

Intravascular Injection

Epidural anesthesia can produce local anesthetic–induced systemic toxicity, primarily through the unintentional administration of drug into an epidural vein. The frequency of vascular puncture by needle or catheter can reach 10%, with rates highest in the obstetric population, in whom these vessels are relatively dilated.[84] Seizures related to epidural anesthesia may be as frequent as 1%.[79] In obstetrics (also see Chapter 33), the likelihood of intravascular injection is decreased by placing the patient in the lateral position during needle and catheter insertion, administering fluid through the epidural needle before catheter insertion, using a single-orifice rather than multiorifice catheter or a wire-embedded polyurethane compared with polyamide epidural catheter, and advancing the catheter less than 6 cm into the epidural space. The paramedian approach, and the use of a smaller-gauge epidural needle or catheter, does not reduce the risk of epidural vein cannulation.

Using epinephrine mixed with local anesthetic as a test dose can be unreliable and so prevention of systemic toxicity should always involve aspiration of the catheter and incremental administration of the local anesthetic.

Subdural Injection

The subdural extra-arachnoid space is easily entered in autopsy attempts in humans. It is an infrequent clinical problem with epidural anesthesia (<1%). Yet, when an epidural block is performed and a higher-than-expected block develops 15 to 30 minutes after injection, subdural placement of local anesthetic must be considered. With a subdural block, the motor block will be modest compared to the extent of the sensory block, and the sympathetic block may be exaggerated. The treatment is symptomatic.

RECENT ADVANCES IN ULTRASONOGRAPHY

Preprocedural ultrasound imaging can accurately identify the intervertebral levels, the midline spinous process, the midline interspinous window, and the paramedian interlaminar window.[49] Imaging of the lumbar spine is significantly easier than that of the thoracic spine, which has narrower interspinous and interlaminar windows. Through these windows the hyperechoic dura (a bright line), the subarachnoid space, and the posterior aspect of the vertebral body may be visualized. Visualization of the ligamentum flavum and epidural space is often more difficult. Ultrasound facilitates identification of the optimal location for needle insertion and an estimation of the skin-to-dura distance. This can be useful in patients with difficult surface anatomic landmarks (e.g., obesity), spine disorders (e.g., scoliosis), or previous spine surgery. Real-time guidance is a highly challenging technique. Ultrasonography in the pediatric population is impressive because of limited ossification of the vertebral column. The epidural catheter tip, dural displacement, and the extent of cranial spread of a fluid bolus can all be visualized.

QUESTIONS OF THE DAY

1. What is the distal termination of the spinal cord in adults and infants? Why is this landmark important to know when performing spinal anesthesia?
2. How does the shape of the spinous processes change from the thoracic to lumbar region? What is the implication for epidural block technique?
3. What is the mechanism of differential sensory block during spinal anesthesia? What is the clinical implication when assessing block height?
4. What are the cardiovascular, respiratory, and gastrointestinal effects of neuraxial anesthesia?
5. What are the absolute and relative contraindications to neuraxial anesthesia?
6. How does local anesthetic baricity impact local anesthetic spread during spinal anesthesia? What patient positions can alter the spread of hyperbaric spinal anesthesia?

7. What local anesthetics are available for spinal use in your institution? How do they differ in terms of duration of action or side effect profile?
8. What is the effect of the following spinal anesthesia additives on quality and/or duration of anesthesia: opioids, vasoconstrictors, alpha agonists?
9. What factors affect epidural anesthesia block height?
10. What are the possible neurologic complications of neuraxial anesthesia? What are the risk factors for each?

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